

## Purpose of This Document

This document describes all data collected by the HearLinX mobile app (Flutter), how it is stored in the backend database (PostgreSQL), and how the local seed database mimics the i-Jejak birth registration system for pilot testing purposes. All data handling is compliant with the Personal Data Protection Act 2010 (PDPA).

## 1. Baby Record Data

Baby records are the foundation of the HearLinX workflow. In production, this data will be pulled automatically from i-Jejak (birth registration system). For pilot testing, baby records are pre-seeded into the local database to mimic i-Jejak.

| Field           | Type             | Required | Source              | Notes                                         |
|-----------------|------------------|----------|---------------------|-----------------------------------------------|
| System ID       | VARCHAR(50)      | YES      | i-Jejak / Seed DB   | e.g. BABY001 — only ID shared externally      |
| Hospital        | UUID (FK)        | YES      | System              | Hospital that registered the baby             |
| Date of Birth   | DATE             | YES      | i-Jejak / Seed DB   | Used for 1-3-6 benchmark tracking             |
| Ward            | VARCHAR(100)     | NO       | Screeener / Seed DB | e.g. Wad Bersalin 3B                          |
| Gestational Age | INTEGER          | NO       | i-Jejak / Seed DB   | In weeks                                      |
| Birth Weight    | INTEGER          | NO       | i-Jejak / Seed DB   | In grams                                      |
| Gender          | CHAR(1)          | NO       | i-Jejak / Seed DB   | M or F                                        |
| Full Name       | TEXT (encrypted) | NO       | i-Jejak / Seed DB   | AES-256 encrypted at rest. Never transmitted. |
| IC Number       | TEXT (encrypted) | NO       | i-Jejak / Seed DB   | AES-256 encrypted at rest. Never transmitted. |

**i-Jejak Integration Note:** In production deployment, baby records will be automatically pulled from i-Jejak (Malaysia birth registration system) at point of delivery. i-Jejak is currently available for Penang hospitals only. For all pilot hospitals, baby records are pre-loaded into the HearLinX database by the coordinator before screening begins. The screener scans the QR wristband to look up the existing record — they do not create new baby records.

## 2. Screening Entry Data (Collected in Flutter App)

This is the primary data collected by the screener at the cot side. Every field entered in the HearLinX mobile app is sent to the backend via encrypted API call (TLS 1.3).

| Field           | Type        | Required | Input Method     | Notes                          |
|-----------------|-------------|----------|------------------|--------------------------------|
| Baby ID         | UUID (FK)   | YES      | QR Scan / Manual | Links screening to baby record |
| Screening Type  | VARCHAR(50) | YES      | Dropdown         | TEOAE, AABR, or ABR            |
| Left Ear Result | ENUM        | YES      | Tap button       | pass or refer                  |

|                  |             |      |                |                                              |
|------------------|-------------|------|----------------|----------------------------------------------|
| Right Ear Result | ENUM        | YES  | Tap button     | pass or refer                                |
| Notes            | TEXT        | NO   | Text input     | Max 100 characters. Optional clinical notes. |
| Attempt Number   | INTEGER     | NO   | Auto / Input   | Default 1. Increments on repeat screening.   |
| Screener ID      | UUID (FK)   | AUTO | From JWT token | Auto-attached by backend from login token    |
| Hospital ID      | UUID (FK)   | AUTO | From JWT token | Auto-attached by backend from login token    |
| Screened At      | TIMESTAMPTZ | AUTO | System         | Timestamp auto-generated by backend          |
| Overall Result   | ENUM        | AUTO | Computed       | pass if both ears pass, else refer           |

### 3. Follow-Up Tracking Data

Automatically created when a REFER result is recorded. Managed by the Hospital Coordinator via web portal or mobile app.

| Field        | Type        | Required | Notes                                                         |
|--------------|-------------|----------|---------------------------------------------------------------|
| Baby ID      | UUID (FK)   | AUTO     | Linked from screening result                                  |
| Screening ID | UUID (FK)   | AUTO     | Linked from REFER screening                                   |
| Status       | ENUM        | YES      | pending → contacted → appointment_booked → closed / escalated |
| Due Date     | DATE        | NO       | Set by coordinator when booking appointment                   |
| Notes        | TEXT        | NO       | Coordinator action notes                                      |
| Created At   | TIMESTAMPTZ | AUTO     | Auto-generated when REFER result is saved                     |
| Updated At   | TIMESTAMPTZ | AUTO     | Auto-updated on every status change                           |

### 4. Local Seed Database — i-Jejak Simulation for Pilot Testing

Since i-Jejak integration is not yet available for all pilot hospitals, HearLinX uses a pre-seeded local database to simulate the data that i-Jejak would provide in production. The seed database is populated by running seed.py before testing begins.

| What is Seeded       | Quantity | Purpose                                   | Mimics                 |
|----------------------|----------|-------------------------------------------|------------------------|
| Test Hospitals       | 1+       | Base unit for all records                 | KKM public hospital    |
| Test Users (4 roles) | 4        | Login credentials for each role           | Real staff accounts    |
| Baby Records         | 10+      | Pre-registered babies for QR scan testing | i-Jejak birth records  |
| Sample Screenings    | 10+      | Mix of PASS and REFER results             | Real cot-side entries  |
| Follow-up Records    | 5+       | Active queue for coordinator testing      | Real follow-up cases   |
| QR Wristband IDs     | 10+      | BABY001–BABY010 scannable system IDs      | Hospital QR wristbands |

**Production Replacement:** When deployed to Penang pilot hospitals, baby records will be pulled live from i-Jejak. For all other hospitals without i-Jejak access, coordinators will manually register babies via the HearLinX web portal before screening. The screener workflow (QR scan → find baby → screen) remains identical regardless of whether data came from i-Jejak or manual registration.

## 5. Data Flow Summary

| Action                     | Where          | API Call                | Result                              |
|----------------------------|----------------|-------------------------|-------------------------------------|
| Screener scans QR          | Flutter App    | GET /babies/{system_id} | Finds baby record in DB             |
| Screener enters result     | Flutter App    | POST /screenings/       | Saves to PostgreSQL via FastAPI     |
| REFER result saved         | Backend (auto) | INSERT into followups   | Follow-up record auto-created       |
| Coordinator views queue    | Web Portal     | GET /followups/         | Reads from PostgreSQL               |
| Coordinator updates status | Web Portal     | PATCH /followups/{id}   | Updates follow-up status in DB      |
| Monthly report generated   | Web Portal     | GET /reports/monthly    | Aggregates from screening data      |
| Excel export               | Web Portal     | GET /reports/export     | Generates .xlsx from DB data        |
| Audit log recorded         | Backend (auto) | INSERT into audit_logs  | Every action timestamped, immutable |

## 6. PDPA 2010 Compliance Notes

| Requirement           | Implementation  | Scope                                                     |
|-----------------------|-----------------|-----------------------------------------------------------|
| Encryption in transit | TLS 1.3         | All API calls between app and server                      |
| Encryption at rest    | AES-256         | Full name and IC number in database                       |
| Data minimisation     | System ID only  | No names or IC numbers transmitted between hospitals      |
| Role-based access     | API-enforced    | Each role sees only what they need — enforced server-side |
| Immutable audit trail | Append-only     | Every action logged with timestamp and user ID            |
| Data residency        | Malaysia region | AWS ap-southeast-1 — no data leaves Malaysia              |
| Baby ID masking       | ****1234        | Shift summary masks baby IDs to last 4 digits             |